1. Project Planning & Management

1.1 Project proposal

Project overview
Many residential buildings in Egypt still manage utilities, tenant communication, and monthly billing manually through paper records, spreadsheets, or messaging apps. This creates errors, delays, lack of transparency, and disputes over shared utility costs.

Project objective
To build a digital platform that helps building administrators manage apartments, residents, utility consumption, alerts, and monthly reporting in a transparent and efficient way.

Project scope

In scope:

1. Building registration and management
2. Apartment and resident management
3. Join-by-access-code workflow
4. Admin approval for resident access
5. Recording global electricity and water meter readings
6. Cost distribution across apartments
7. Alert system
8. Monthly PDF report generation
9. Mobile app with offline cache

Out of scope

1. Smart meter hardware integration in phase 1
2. Online rent payment gateway
3. National ID verification integration
4. Multi-city government integrations
5. IoT automation beyond manual/semimanual data entry

1.2 Problem statement

Residential building management in Egypt often lacks a unified digital system for tracking residents, utility usage, shared costs, and emergency communication. This results in inefficient administration, poor accountability, and limited visibility for residents.

1.3 Goals

- Reduce manual work for building administrators
- Improve transparency of utility cost allocation
- Enable secure resident onboarding
- Support timely alert communication
- Provide reliable monthly documentation through PDF reports
- Maintain usability even with weak internet through offline caching

1.4 Deliverables

- Flutter mobile application
- Supabase backend and PostgreSQL schema
- SQLite offline cache layer
- Technical documentation
- API/database documentation
- Test cases and QA report
- Presentation slides
- User manual
- Source code on GitHub

1.5 Milestones

Phase	Deliverables
Project Planning & Management	Proposal, plan, roles, risks, KPIs
Literature Review	Related systems, methods, justification
Requirements Gathering	Stakeholders, user stories, requirements
System Analysis & Design	UML, ERD, DFD, architecture, UI/UX
Implementation	Code, repository, CI/CD, deployment
Final Presentation & Reports	User manual, technical report, presentation

1.6 Team roles

Role	Responsibilities
Project Manager	planning, timeline, risk tracking, coordination
Flutter Developer	mobile UI, state management, local storage
Backend Developer	Supabase integration, database, auth, APIs
Database Designer	schema, constraints, optimization
QA Engineer	testing plan, test cases, bug tracking
UI/UX Designer	wireframes, mockups, usability
Documentation Lead	report writing, GitHub docs, presentation

1.7 Risk assessment and mitigation

Risk	Impact	Mitigation
Internet instability	app may fail to sync	use SQLite offline cache and sync queue
Incorrect meter entry	billing disputes	validation rules, admin review, edit logs
Security/privacy issues	exposure of resident data	row-level access control, secure auth, audit logs

Scope creep	delays	define MVP features and freeze scope per milestone
PDF generation issues	poor reporting	standard templates and preview testing
Limited user adoption	weak project success	simple UX, Arabic-friendly design, onboarding guide

1.8 KPIs

- App response time under 2–3 seconds for common actions
- Data sync success rate above 95%
- Monthly report generation success rate above 99%
- Reduction in billing disputes
- User adoption rate by apartments in a building
- Alert delivery/read rate
- Admin task completion time vs manual process